

Applied Data Driven Methods in Engineering – Project

One group submits a single report. Show plots, data used and codes. Complete by Saturday 6th December, 5 PM

1. The DMD Code given first generates a data. DMD is used to analyze the data. It creates the mapping matrix $X'=Ax$, finds the dominant modes and time evolution metrics, and finally re-generates the data.

Assume a data of your own. Use DMD on the data to show the dominant modes, comment on time-evolution metrics and regenerate your data.

2. Get a time varying data of a single variable from internet.

(i) Obtain the frequency – amplitude, DFT plot.

(ii) show implementation of a low-pass filter, high-pass filter and band-pass filter of your choosing on the same data. Compare filtered results with original, both in frequency and time domain.

3. Choose a discrete time data $x[n]$ (atleast 8 non-zero elements). Plot,

(i) $x[n]$ (ii) $x[n-2]$ (iii) $x[n+1]$

(iv) $y[n] = 0.2 * x[n] + 0.3 * x[n - 1] + 0.2 * x[n - 2]$ - compare DFT of $x[n]$ and $y[n]$

(v) $y[n] = 0.2 * x[n] + 0.3 * x[n - 1] + 0.5 * y[n - 1]$ (take $y(-1)=0$). - compare DFT of $x[n]$ and $y[n]$

4. Attached SINDy code first numerically solves three 1st order differential equations of a Lorenz system, “EX02_Lorenz.m”. The differential equations for Lorenz system is defined in “./utils/Lorenz.m”. System parameters are assumed (sigma, beta, alpha).

These solutions are taken as input. A library considering terms upto 3rd degree is created. Sindy is implemented to identify the dominant terms in the system equations and the associated parameter.

Choose a system / differential equation of your own (atleast 2 state-variables). Solve the system using “ode45” (Line 11-27 in EX02_Lorenz.m”. Apply SINDy to the solutions obtained to identify the system and parameters. Show results.

5. Choose $x[n]$ and $y[n]$ of your own. Find their convolution $x[n]*y[n]$, cross-correlation of $x[n]$ with $y[n]$. (use in-built command)

6. Create a matrix A of size 6X4, 6 rows and 4 columns. Obtain the SVD of the matrix. Create a matrix approximation of A considering the 1st two dominant modes.

7. Take a data of your choosing. Use polynomials ($1, t, t^2, t^3, \dots$) or use sine and cosine series of appropriate terms to best fit the data, Use least squares regression, compute the pseudo-inverse, generate the best-fit line and compare against original data.

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